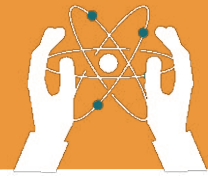


UNIT 2

SENSE ORGANS

Learning objectives:

- Understand the functions of the five sense organs: eyes, ears, nose, tongue, and skin.
- Identify key parts of the eyes, nose, and ears and their role in detecting light, smell and sound.
- Recognize the role of skin receptors in sensing touch, temperature, and pain.



SENSE ORGANS

Sense Organs

Sense organs help us interact with the things around us. They detect different types of stimuli like light, sound, and heat. The five major sense organs are the eyes, ears, nose, tongue, and skin.

Eyes

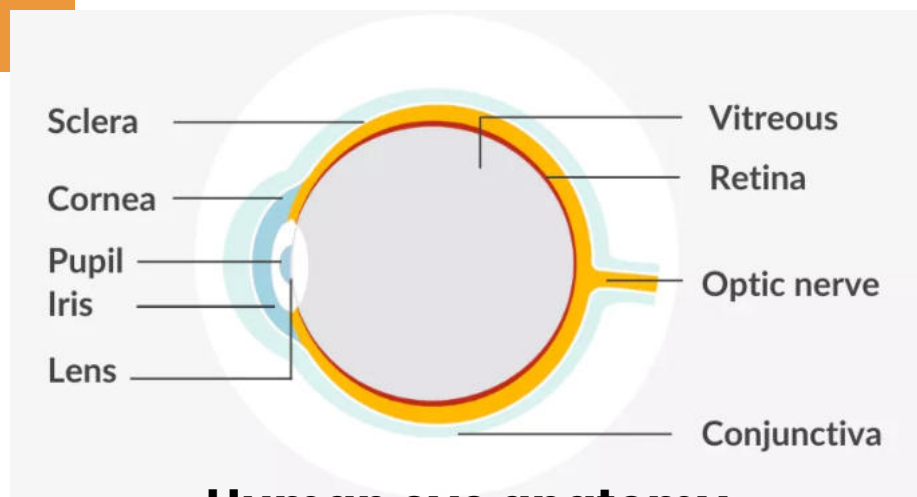
Function: The eyes allow us to visualize things around us by detecting colour, light and movement.

Eyes are complex organ with several important parts.

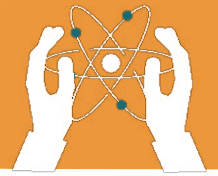
Parts of the Eye:

- **Cornea:** The clear, outer part that focuses light.
- **Iris:** The coloured part that controls the size of the pupil.
- **Pupil:** The opening that lets light enter the eye.
- **Lens:** It focuses light onto the retina.
- **Retina:** It contains cells that detect light and send signals to the brain.
- **Optic Nerve:** It carries signals from the retina to the brain.

How the Eye Works: Light enters the eye through the cornea, passes through the pupil, and is focused by the lens onto the retina. The retina converts light into signals sent to the brain, forming images.



Human eye anatomy



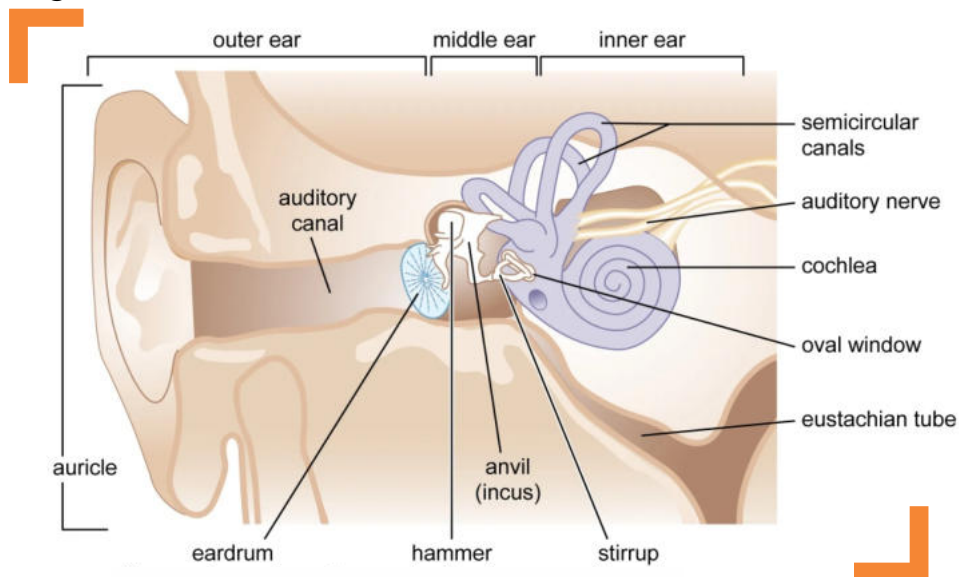
Ears

Function: Ears are the sensory organs responsible for hearing. The ears allow us to hear by detecting sound waves.

Parts of the Ear:

- **Outer Ear:** It collects sound waves and directs them into the ear canal.
- **Middle Ear:** It contains the eardrum and three tiny bones (malleus, incus, and stapes) that amplify sound.
- **Inner Ear:** It contains the cochlea, which converts sound vibrations into electrical signals that are sent to the brain.

How the Ear Works: Sound waves enter the outer ear, vibrate the eardrum, and pass through the middle ear bones. The inner ear changes these vibrations into signals that travel to the brain, allowing us to hear.

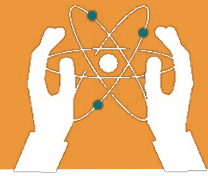


Human ear anatomy

• DID YOU KNOW?

The three tiny bones in the middle ear—malleus, incus, and stapes—are the smallest bones in the human body and are essential for hearing.





SENSE ORGANS

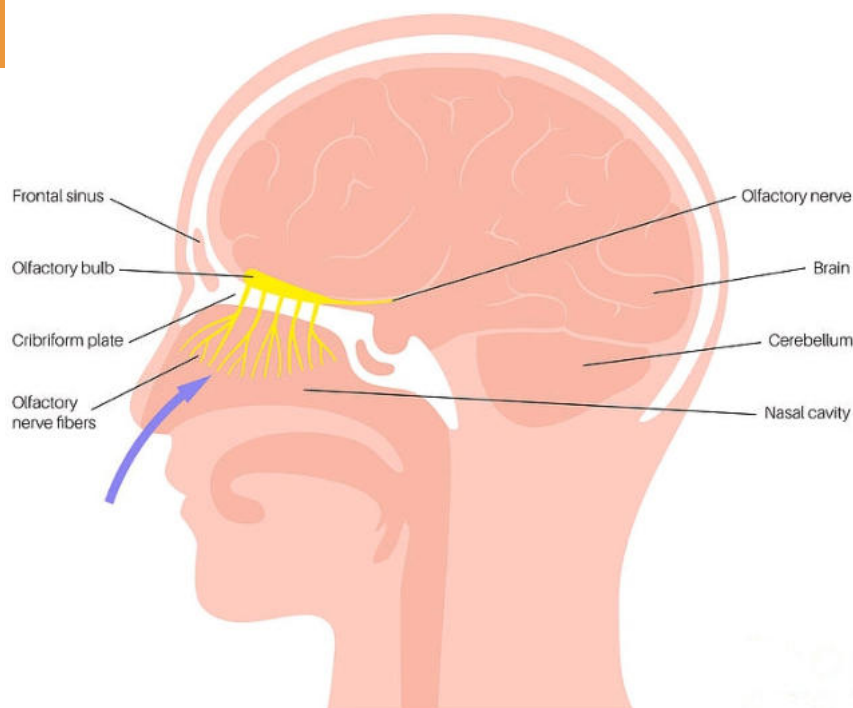
Nose

Function: Nose is the sensory organ for detecting odour around us. The nose helps us smell by detecting chemicals in the air.

Parts of the Nose:

- **Nasal Cavity:** It contains odor receptors sensitive to smell.
- **Olfactory Nerve:** It transmits smell signals from the nose to the brain.

How the Nose Works: When chemicals in the air enter the nasal cavity, they stimulate the receptors. These signals travel through the olfactory nerve to the brain, where they are interpreted as different smells.

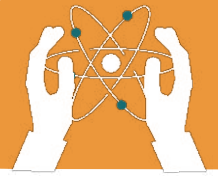


Parts of the Nose

• DID YOU KNOW?



Nose can detect over 1 trillion different scents.



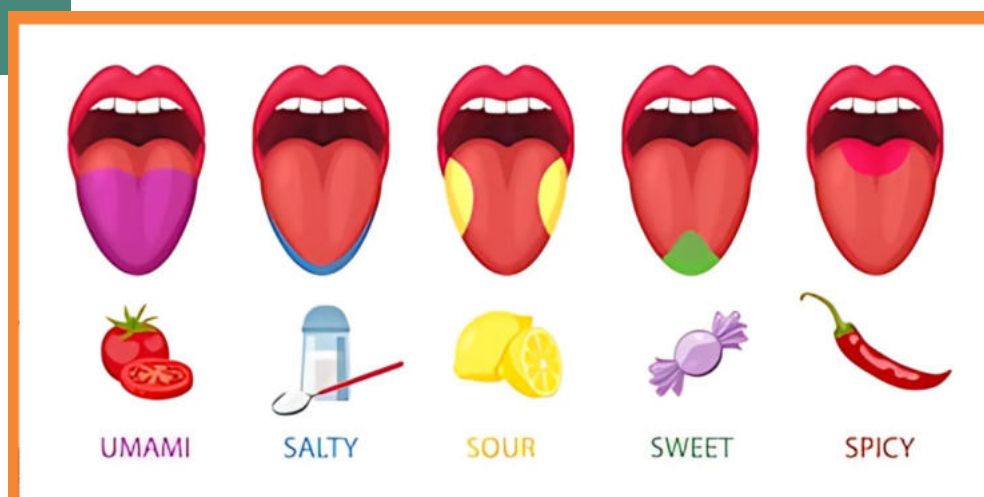
Tongue

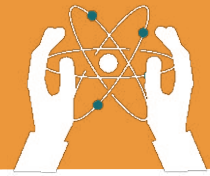
Function: Tongue is a sensory organ use to taste things. The tongue allows us to chew, swallow, speak etc.

Parts of the Tongue:

- **Taste Receptors:** The tongue has taste buds that detect sweet, sour, salty, bitter, and umami flavours.
- **Structure of the Tongue:** The surface of the tongue is covered in tiny bumps called papillae, which contain taste buds.

How the Tongue Works: When we eat, food particles dissolve and stimulate the taste buds. These taste buds send signals to the brain, allowing us to taste different flavours.





SENSE ORGANS

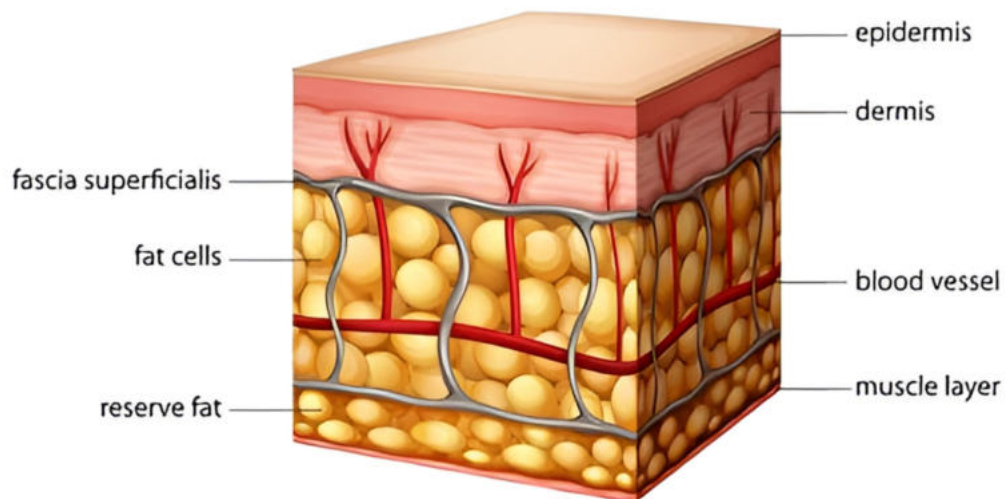
Skin

Function: The skin senses touch, temperature, and pain.

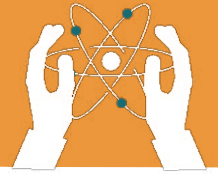
Structure: The skin has three layers: the epidermis, dermis, and hypodermis. It contains receptors for different types of sensations.

How the Skin Works: Receptors in the skin detect touch, temperature, and pain. These receptors send signals to the brain, helping us respond to our environment.

Anatomy of Human Skin



SENSE ORGANS



Give reasons for the following.

1. Why do we have five sense organs?

2. Why does food taste different when we have a cold?

Answer the Following Questions:

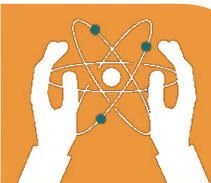
Short-Answer Questions

Q1. What are sense organs?

Q2. How many sense organs do humans have?

Q3. What is the structure of the skin and how does it help protect the body?

Q4. What is the structure of the nose and how does it help the body?

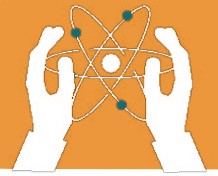


SENSE ORGANS

Long-Answer Questions.

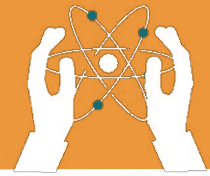
- Q1. Describe the structure and function of the eye. How does it help us to see?
- Q2. Explain how the ear helps us hear. Describe its main parts and their functions.
- Q3. Draw neat and labelled diagram of human ear with labelling.
- Q4. Explain the structure and function of the human ear and describe how it helps us to hear.

SENSE ORGANS



Fill in the blanks.

1. The _____ controls the amount of light entering the eye.
2. The _____ is the part of the ear that vibrates when sound waves hit it.
3. The _____ nerve carries smell signals from the nose to the brain.
4. Taste buds on the tongue can detect _____, _____, _____, _____, and _____ flavours.
5. The _____ layer of the skin contains receptors for touch and pain.



SENSE ORGANS

Write “T” for True or “F” for False.

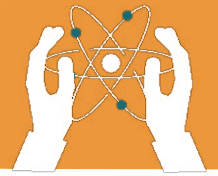
1. The lens focuses light on the retina to help us see.

2. The outer ear contains the cochlea.

3. The nose detects smells through chemicals present in the air.

4. Taste buds are found on the tongue and help detect sound.

5. The skin contains receptors that can sense pain.



Choose the correct answer.

1. Which part of the eye is responsible for focusing light?

- Retina
 - Iris
 - Lens
 - Cornea
-

2. The cochlea is part of which sense organ?

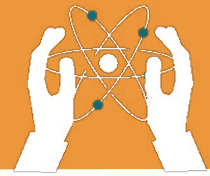
- Eye
 - Ear
 - Nose
 - Skin
-

3. The nasal cavity is related to the sense of:

- Hearing
 - Smell
 - Sight
 - Touch
-

4. Which sense organ has receptors for touch, temperature, and pain?

- Nose
 - Skin
 - Tongue
 - Eye
-



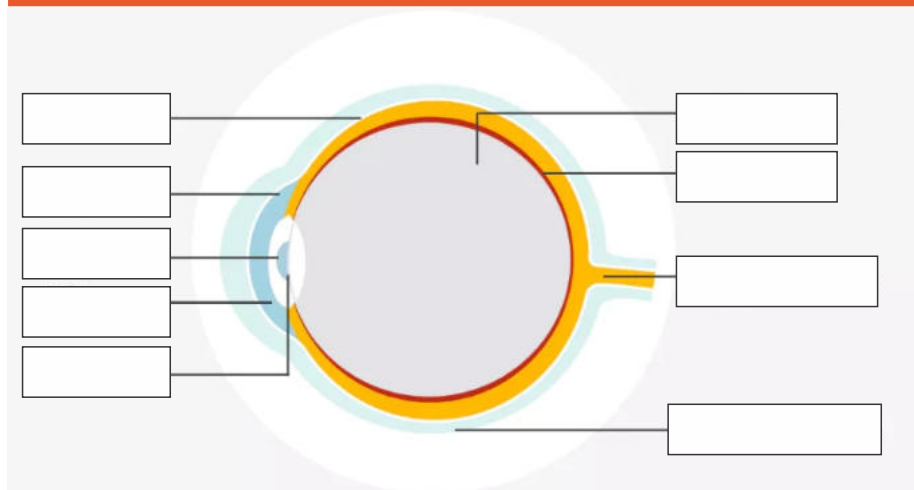
SENSE ORGANS

5. The taste buds on the tongue can detect _____ flavours.

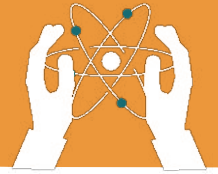
- Three
 - Five
 - Two
 - Six
-

Label the diagram :

Human eye anatomy



SENSE ORGANS



Match the Column.

COLUMN A

Retina

Cochlea

Olfactory Nerve

Taste Buds

Skin Receptors

COLUMN B

Detects smell

Detects taste

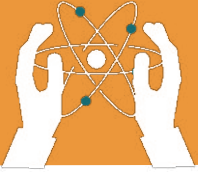
Detects sound

Detects touch

Detects light

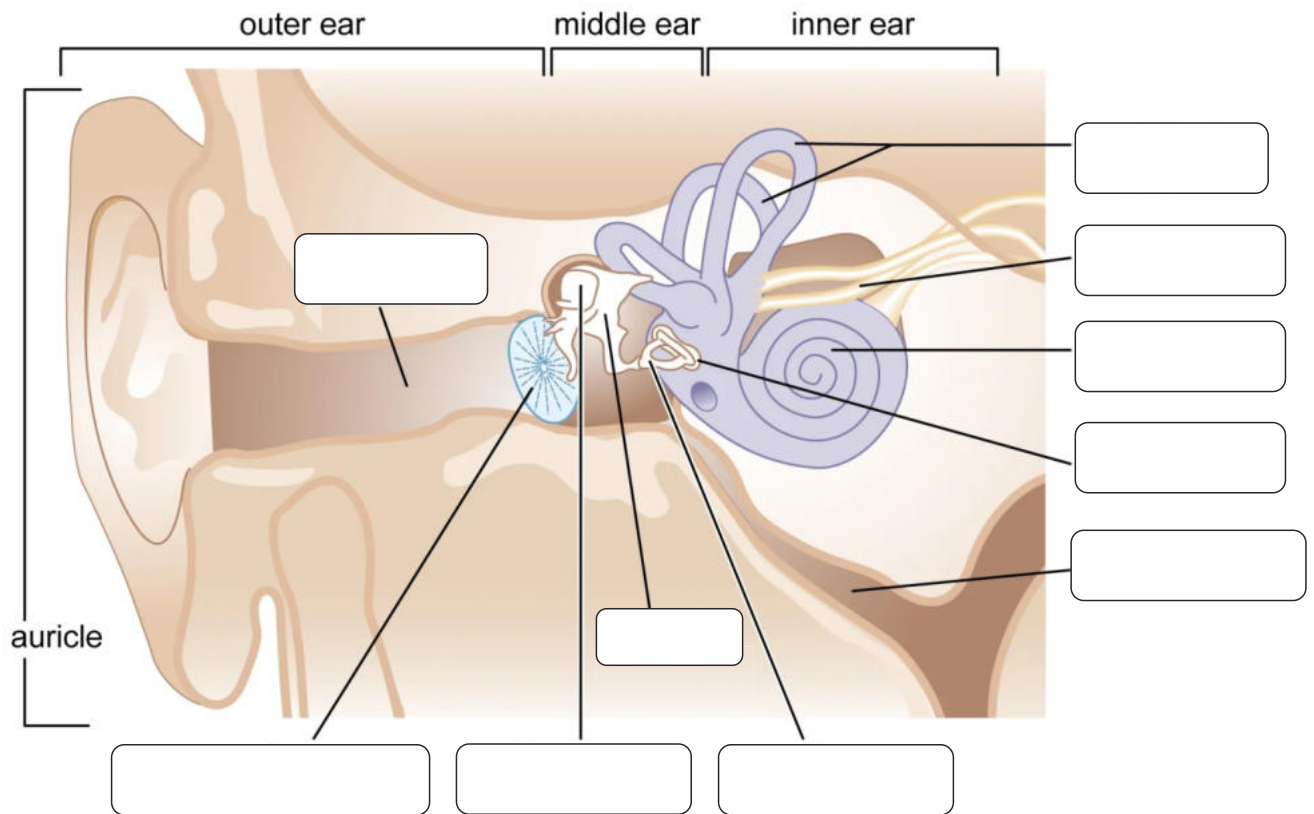
Project Idea

Sense Exploration Project: In groups, students explore each sense with materials (flashlight, tuning fork, scented items, flavoured snacks, textured objects) and record observations on how each sense detects stimuli.

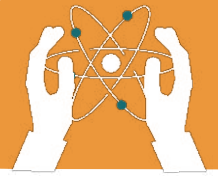


SENSE ORGANS

Label the diagram of a Human Ear.



SENSE ORGANS



"Sense Organs in Action" Presentation



ACTIVITY

1. Objective: In groups, students research and create a model of one sense organ (eye, ear, nose, tongue, or skin), showing its structure and function.

2. Preparation: Each group builds a labelled model, prepares a 5-minute presentation on how the organ works, and shares fun facts.

3. Presentation Day: Groups present their model, explaining the organ's function and answering classmates' questions.

Reflection: Students write in their journals about what they learned from each presentation and how sense organs help us interact with the world.